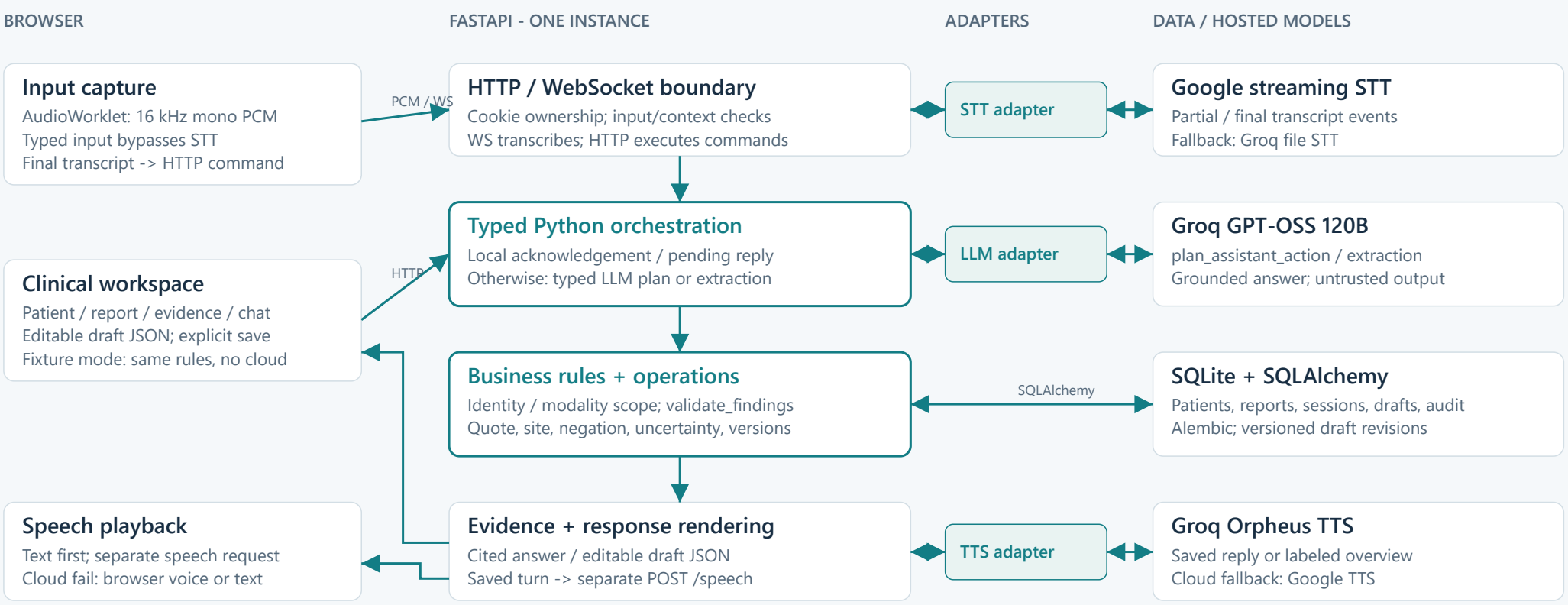
VI-SCAN | Architecture and report-generation flows

Logical components above; numbered execution paths below. Synthetic-data proof of concept - no imaging diagnosis or clinical signing.



LLM interprets and selects evidence; Python resolves identity, queries records and checks draft evidence and qualifiers before saving.

All cloud links pass through adapters for timeouts, retries and fallback. Keys stay server-side. Exact replies / pure thanks can stay local.

RETRIEVAL

1 Voice: STT / typed text -> 2 HTTP command -> 3 retrieve plan -> 4 identity + ambiguity rules -> 5 latest-report SQL -> 6 report + source

DICTATION TO DRAFT

1 Transcript -> 2 LLM extraction -> 3 validate_findings (quotes, site, negation, uncertainty) -> 4 editable JSON -> 5 clinician review -> 6 versioned save

CONTEXTUAL FOLLOW-UP

1 Question -> 2 session patient / report scope -> 3 grounded LLM evidence -> 4 quote / version checks -> 5 answer + source excerpt -> optional TTS

293 local tests | Synthetic data only | One worker; hosted inference | Historical voice evidence; current quota not guaranteed